



Mastering the Art of Penetration Testing

A Step-by-Step Guide to Finding
and Fixing Cybersecurity
Vulnerabilities

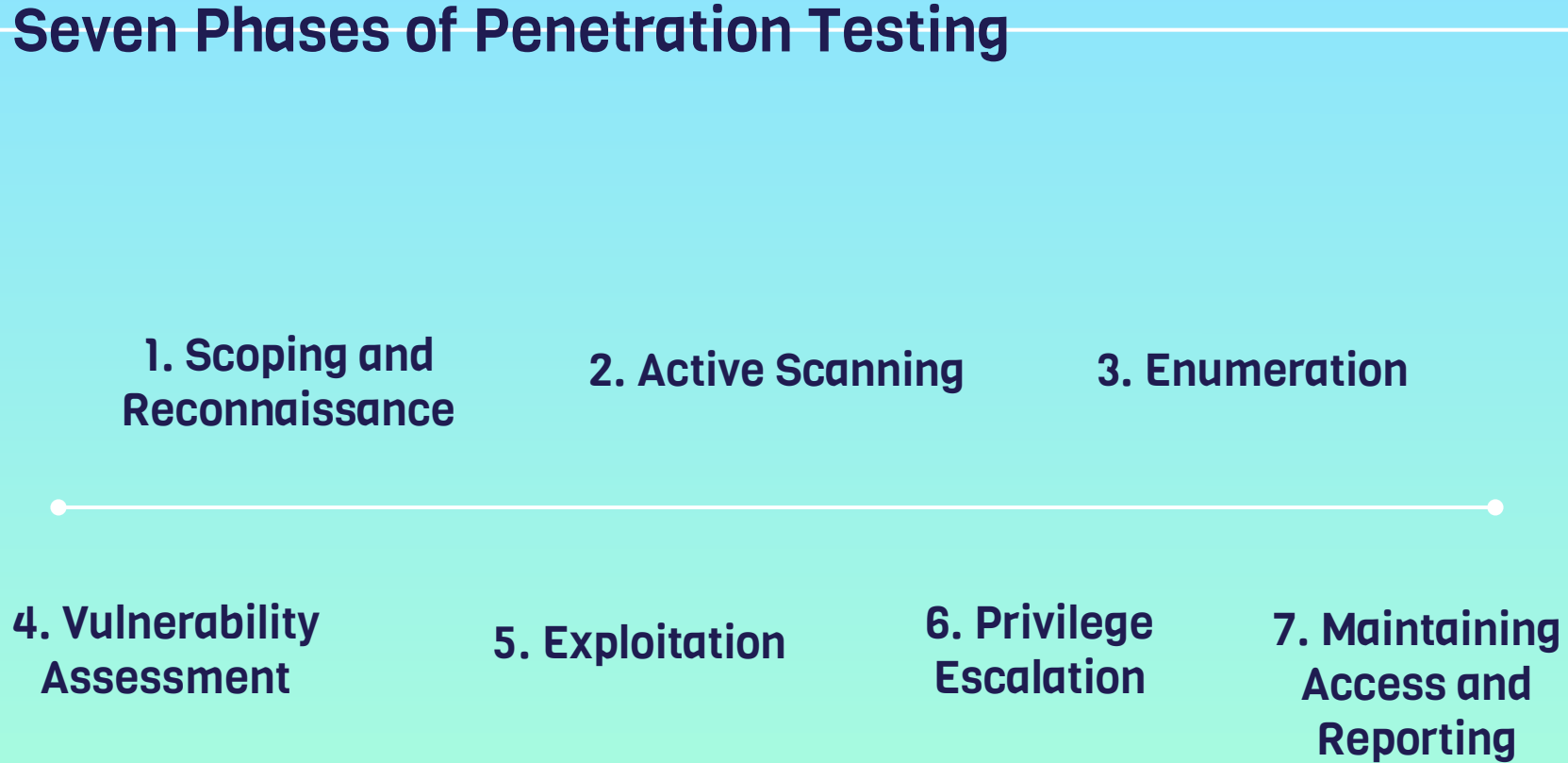
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What is Penetration Testing?



- A security exercise that simulates an attack in an attempt to find and exploit vulnerabilities in a system.
- Helps to identify any weak spots in a system's defenses that could potentially be taken advantage of.

Seven Phases of Penetration Testing



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graph LR; 1[1. Scoping and Reconnaissance] --> 2[2. Active Scanning]; 2 --> 3[3. Enumeration]; 3 --> 4[4. Vulnerability Assessment]; 4 --> 5[5. Exploitation]; 5 --> 6[6. Privilege Escalation]; 6 --> 7[7. Maintaining Access and Reporting];
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**1. Scoping and
Reconnaissance**

2. Active Scanning

3. Enumeration

**4. Vulnerability
Assessment**

5. Exploitation

**6. Privilege
Escalation**

**7. Maintaining
Access and
Reporting**

Scoping and Reconnaissance

Goals

- Define the scope
- Gather data

Tools Used

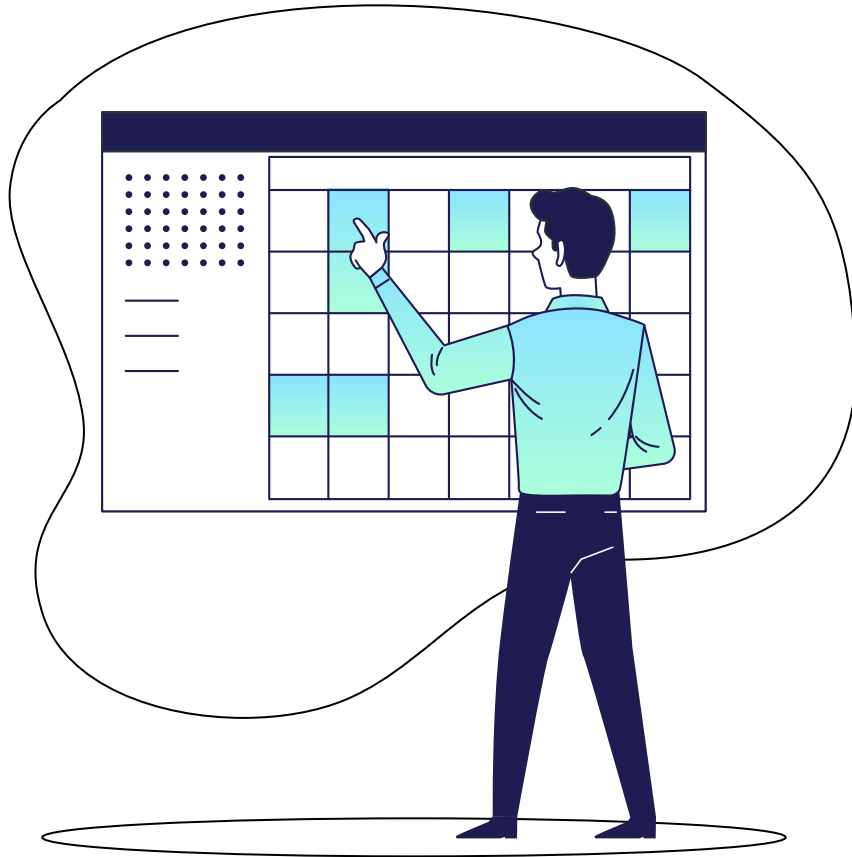
- Nmap
- Shodan
- Inteltechniques
- Pimeyes

Key Tasks

- Identify infrastructure, users, and boundaries



Active Scanning



Goals

- Detect live hosts and open ports

Tools

- Nmap
- Shodan

Output

- IP Addresses
- Port states
- Services

Enumeration

Goals

- Extract service details

Tools

- Nmap
- Inteltechniques

Focus

- Software versions
- Configurations



Vulnerability Assessment



Goals

- Identify known vulnerabilities

Tools

- Nessus
- OpenVAS

Output

- CVEs
- Risk levels

Exploitation

Goals

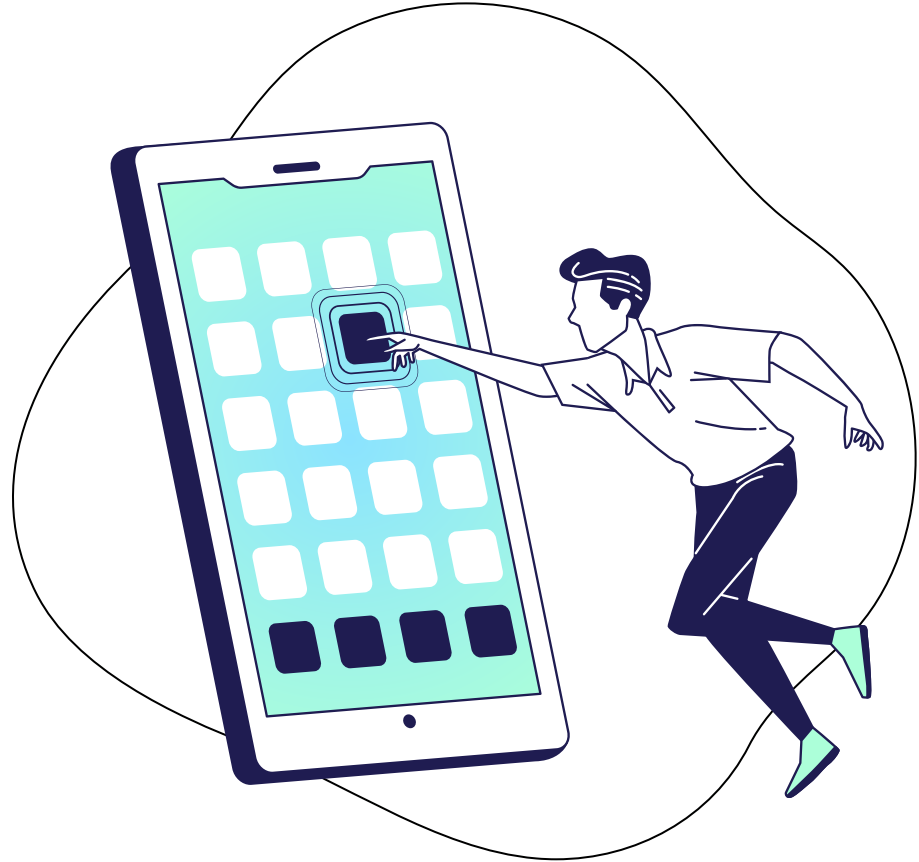
- Simulate real attacks

Tools

- Metasploit

Outcome

- Unauthorized access (still within scope)



Privilege Escalation



Goals

- Gain administrative access

Tools

- Mimikatz

Action

- Extract credentials
- Elevate access

Maintaining Access and Reporting

Goals

- Simulate persistence

Tools

- Cobalt Strike

Outcome

- Final report with findings and fixes



Conclusion



Proactive

Penetration testing is a proactive form of defense



Continuous

Continuous testing leads to stronger systems



Document

Make sure everything is documented and reported